

Perfectoidness of moduli spaces of quotient bundles on FF curve.

$F \subset \mathbb{Q}_p$ fin. extn. with residue field $\overline{\mathbb{F}_q}$. C : complete alg. closed field containing $\overline{\mathbb{F}_q}$.

C^b : tilt. Perf_{C^b} : perfectoid spaces / C^b . $T \in \text{Perf}_{C^b}$. $X_T := X_{\text{FF}, T}$ relative Fargues-Fontaine curve.

Recall that we have an equiv. $\text{Perf}_C \cong \text{Perf}_{C^b}$. We denote the bundle $\mathcal{O}(\frac{a}{b})$ with rank and degree a , for a reduced fraction $\frac{a}{b}$. On a geometric FF curve, every vector bundle can be written as a $\oplus$ of some v.b. $\mathcal{O}(\lambda_i)$ for $\lambda_i \in \mathbb{Q}$, reduced. In this note, a subbundle $E' \hookrightarrow E$ means that E/E' is also a v.b. and every semistable vector bundle can be written as

[FF Theorem 8.2.10] Classification of vector bundle / X_{FF} .

$\mathcal{O}(\lambda)^{\oplus m}$ for some $\lambda \in \mathbb{Q}$, $m \in \mathbb{Z}_+$.

(Chai's Defn. 5.1)

Hansen-Mann's moduli problem: For $0 < d < n$, the moduli functor

$W_{n,d}: T/C^b \mapsto$ isomorphism class of the quotient

$q: E_n|_{X_T} \rightarrow E$ where $E|_{\bar{t}} \cong E_d$ for every geom point $\bar{t}$.

Then $W_{n,d}$ is a v-sheaf.

Theorem 1 $W_{n,d}$ is represented by a perfectoid space.

We first prove two lemmas.

Lemma 2. Let S be a perfectoid space of char p . Q : v.b. on X_S . Let $s \in H^0(X_S, Q)$ be a global section. Then the functor

$$\mathcal{Z}_s: T \mapsto \{f: T \rightarrow S \mid f_x^* s = 0 \text{ on } X_T\} \quad \begin{aligned} f: T \rightarrow S &\mapsto f_x: X_T \rightarrow X_S \\ &\mapsto f_x^*: H^0(X_S) \rightarrow H^0(X_T). \end{aligned}$$

is represented by a Zariski closed perfectoid subspace of S .

[FST II.2.6, [KL] 6.2.4).

Proof: The statement is analytically local, so we may assume $S = \text{Spa}(R, R^\dagger)$ is perfectoid.

Recall the construction of FF curve $X_{\text{FF}} = X_{\text{FF}}/p^{\mathbb{Z}}$. $\mathbb{Z}$ a slope k on Y_{FF} , we just need to consider this problem on $Y_S[[1, q]]$. Since the Frobenius φ translates over Y_S . Let $B := \mathcal{O}(Y_S[[1, q]])$.

Let $p: Y_S \rightarrow Y_S/p^{\mathbb{Z}} = X_S$. $j: Y_S[[1, q]] \hookrightarrow Y_S$. $Q[[1, q]] := j^* p^* Q$. Since $Y_S[[1, q]]$ is affinoid

hens-perfectoid, there exists an equiv. between {v.b. / $Y_S[[1, q]]$ } and finite projective B -modules.

Thus $Q[[1, q]]$ corresponds to a finite projective B -module $M = H^0(Y_S[[1, q]], Q[[1, q]])$.

Moreover, there exists a finite projective B -module M' s.t. $M \oplus M' \cong B^{\oplus k}$ for some $k \in \mathbb{Z}_+$.

equivalently. $\exists$ an injection $M \hookrightarrow B^k$. Let $S_{[1,q]} := j^* p^* S \in M$. then it can be represented by a tuple $(f_1, \dots, f_k) \in B^k$.

Let $f: T \rightarrow S$ be a morphism of perfectoid spaces. then the base change $M_T \hookrightarrow B_T^k$ is still split. where $B_T = H^0(Y_T, \mathcal{O}_{Y_T} \otimes \mathcal{O}_{[1,q]})$. M_T : a finite projective B_T -module. Let $S_{[1,q],T} \in M_T$ be the image of $S_{[1,q]}$, with coordinates $(f_{1,T}, \dots, f_{k,T}) \in B_T^k$. then $S_{[1,q],T} = 0 \Leftrightarrow (f_{1,T}, \dots, f_{k,T}) = (0, \dots, 0)$. Thus $S_T = 0 \Leftrightarrow S_{[1,q],T} = 0 \Leftrightarrow f_{1,T} = \dots = f_{k,T} = 0$.

Let $J := (f_1, \dots, f_k) \in B$ be the ideal generated by f_i . then $f_{i,T} = 0 \Leftrightarrow$ the ideal J becomes 0 under the ring map $B \rightarrow B_T$. Thus $S_T = 0 \Leftrightarrow J \cdot B_T = 0$. Then by [FS, IV. 4.3] for the finitely generated ideal $J \subseteq B$. $\exists$ a Zariski closed perfectoid subspace $Z \subseteq S$. satisfying the following property: for $\forall T \rightarrow S$. $T \rightarrow Z \Leftrightarrow J \mapsto 0$ in B_T . In other words, $Z(T) = \{T \rightarrow S \mid J \mapsto 0\}$. Thus Z_S is presented by Z . □

Lemma 3. A : semistable v.b. on the geometric Fargues-Fontaine curve X_{cb} . $\text{rank } A = h \geq 1$. and $\deg A = a < h$. Let $C = h - a$. $N = h + 1$. then there exists a seq. of vector bundles.

$$0 \rightarrow A \xrightarrow{\iota} B_N = \mathcal{O}\left(\frac{h}{h+1}\right) \xrightarrow{\pi} \mathcal{O}(C) \rightarrow 0.$$

Proof: Recall [BFHT, Theorem 1.1.2], if F_1, F_2 are semistable with slopes $\mu(F_1) \leq \mu(F_2)$. then a bundle B fits in an extn $0 \rightarrow F_1 \rightarrow B \rightarrow F_2 \rightarrow 0 \Leftrightarrow$ the Harder-Narasimhan polygons

$$\text{HN}(B) \subseteq \text{HN}(F_1 \oplus F_2).$$

Applying this to $F_1 = A$, $F_2 = \mathcal{O}(C)$, $B = B_N$. Note that.

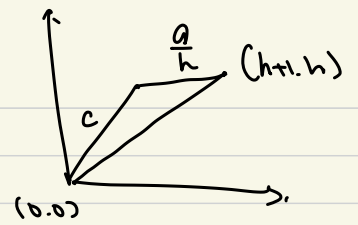
$$\text{rank}(A \oplus \mathcal{O}(C)) = h + 1 = \text{rank } B_N.$$

$$\deg(A \oplus \mathcal{O}(C)) = a + C = h = \deg B_N.$$

Since $a < h$ and $a \in \mathbb{Z}$. $a \leq h-1$. hence

$$\frac{a}{h} \leq \frac{h-1}{h} < 1 \leq C. \Rightarrow \mu(A) < \mu(\mathcal{O}(C)).$$

Since the bundle B_N is stable, its HN polygon is a straight line, and the HN polygon of $A \oplus \mathcal{O}(c)$ has one slope equal to $\frac{a}{h}$ and one larger slope equal to c . Since the two bundles have the same deg and rank, HN polygon of $A \oplus \mathcal{O}(c)$ lies above the straight polygon of B_N . Therefore we have the extn



$$0 \rightarrow A \rightarrow B_N \rightarrow \mathcal{O}(c) \rightarrow 0.$$

□.

Choose one such exact seq. once and for all. Since $\mathcal{O}(c)$ is a v.b., the seq. is locally split. Thus for any perfectoid space T , after pulling back to X_T , the seq. is still exact

$$0 \rightarrow A_T \rightarrow B_{N,T} \rightarrow \mathcal{O}(c)_T \rightarrow 0.$$

Moreover, in our convention, $\mathcal{O}(c) \otimes A$ is a subbundle of B_N , since $\mathcal{O}(c)$ is a v.b. again.

Let's begin to prove our main theorem. Set $r = n-d$, $V = E_n$, $A := \wedge^r E_n$, then $h := \text{rank } A = \binom{n}{r}$, $a := \deg A = \binom{n-1}{r-1}$, thus $\mu(A) = \frac{\deg A}{\text{rank } A} = \frac{r}{n} < 1$. Let $c := h - a = \binom{n-1}{r}$, $N := h+1$, then by **Lemma 3**, we may choose once and for all an exact sequence

$$0 \rightarrow A \xrightarrow{i} B_N \xrightarrow{\pi} \mathcal{O}(c) \rightarrow 0, \quad \text{where } B_N = \mathcal{O}\left(\frac{h}{h+1}\right).$$

Our goal is to realize $W_{n,d}$ as a closed subspace of the known space $W_{N,1}$. By definition, a point in $W_{N,1}$ is a quotient $E_N \rightarrow D$, where $D_F \cong \mathcal{O}(1)$ on geometric point F . Taking dual and twisting by $\mathcal{O}(1)$, we get $D^\vee(1) \hookrightarrow E_N^\vee(1)$. Define $L := D^\vee(1)$, and

$$E_N^\vee(1) = \mathcal{O}\left(\frac{N-1}{N}\right) = \mathcal{O}\left(\frac{h+1-1}{h+1}\right) = \mathcal{O}\left(\frac{h}{h+1}\right) = B_N.$$

Thus each point in $W_{N,1}$ gives a universal subbundle $L \hookrightarrow B_N$. Then $L \subseteq A \Leftrightarrow$ the composition

$$L \xrightarrow{i} B_N \xrightarrow{\pi} \mathcal{O}(c) \quad \text{with a unique map } \psi_A: L \rightarrow A \text{ s.t. } i \circ \psi_A = \psi.$$

Note: Here we need $W_{N,1}$ to be perfectoid.

this is done by [JLH].

is zero. By **Lemma 2**, its vanishing defines a Zariski closed perfectoid subspace $Z_i \hookrightarrow Q$, parametrizing lines $L \subseteq B_N$ which actually lie in $A = \wedge^r V$. By the universal property of u , we know that $L \subseteq B_N$ is a real subbundle.

However, a line $L \subseteq \wedge^r V$ doesn't automatically come from a rank r subbundle K of V , i.e. $L = \det K$. For this to happen, the vectors in L must be decomposable.

Recall the Plücker embedding $\text{Gr}(r, V) \hookrightarrow \mathbb{P}(\wedge^r V)$. The image is cut out by a system of quadratic equations

$$(L_f z) \wedge z = 0$$

for $\forall f \in \wedge^{r-1} V^*, z \in \wedge^r V$. $(r-1)$ -form f , and L_f is the contraction. Locally choose a generator $z \in \wedge^r V$ of L . They therefore combine into a global section p of the vector bundle

$$R := L^{-2} \otimes \wedge^{r-1} V \otimes \wedge^{n-r+1} V.$$

Then $p=0$ corresponds to a Zariski closed perfectoid subspace

$$Z_2 \hookrightarrow Z_1$$

by **lemma 2**, which represents those lines in $\wedge^r V$ satisfying the Plücker condition.

Then a decomposable line $L \subseteq \wedge^r V$ corresponds uniquely to a rank r subbundle $K \subseteq V$.

s.t. $\det K = L$. Let $E := V/K$. Claim: $E \cong E_d$. Thus each point in Z_2 gives a point in Wd.n.

Since L is a line bundle, L is locally trivial on the geometric points, thus $\deg L = 0$. Then $\deg K = \deg(\det K) = \deg L = 0$. For any proper nonzero subbundle $H \subseteq K$, since $K \subseteq V = E_n$, $H \subseteq E_n$. E_n is stable, with slope $\frac{1}{n}$, thus $\mu(H) < \frac{1}{n}$, $\deg H < \frac{\text{rank } H}{n} < \frac{n}{n} = 1$.
 $\Rightarrow \deg H \leq 0$, $\mu(K) = 0$. Thus $\mu(H) \leq 0 = \mu(K)$. Then K is semistable of slope 0.

By the classification theorem, $K \cong \mathcal{O}^r$ for some integer r .

$$\text{rank } E = \text{rank } V - \text{rank } K = n - r = n - (nd) = d, \deg E = \deg V - \deg K = 1$$

Now it suffices to show that E is semistable. Note that all HN slopes of $E \geq 0$

If not, suppose the last HN quotient Q has slope $\mu(Q) \leq 0$, then the composition

$V \twoheadrightarrow E \twoheadrightarrow Q$ is a nonzero map. Since $V = E_n$ semistable with slope $\frac{1}{n} > 0$,

$\mu(Q) \leq 0$, $\text{Hom}(V, Q) = 0$, a contradiction. Thus each HN factor E_i has positive slope, $\deg E_i > 0 \Rightarrow \deg E_i \geq 1$. But $\sum_i \deg E_i = \deg E = 1 \Rightarrow$ unique factor.

E is semistable. By classification, $E \cong \mathcal{O}(\frac{1}{d}) = E_d$. Thus we have a map.

$$Z_2 \rightarrow \text{Wd.n.}$$

On the other hand, we will show that this map is an isomorphism actually.

For a perfectoid space T , let $V_T := E_n|_{X_T}$. A $W_{n,d}(T)$ -point gives a c.s.

$$0 \rightarrow K \rightarrow V_T \rightarrow E \rightarrow 0.$$

Let $L = \det K$. Then $L = \det K \hookrightarrow \wedge^n V_T = A_T$. Using the fixed extn. $A_T \hookrightarrow B_{n,T}$, we have

$L \hookrightarrow B_{n,T}$. Dualizing and twisting by $\mathcal{O}(1)$, of the line subbundle $L \hookrightarrow B_{n,T}$, we get $E_N \rightarrow L^{\vee}(1)$.

$L^{\vee}(1)$ is locally trivial on the geometric point. Thus this gives a $W_{n,1}(T)$ -point.

and it belongs to $\mathbb{Z}_2$ b/c $L \hookrightarrow A_T \Rightarrow L \hookrightarrow B_T \rightarrow \mathcal{O}(1)$ is zero and $L = \det K$. Both constructions are

compatible with base change. We get an isom. of v -sheaves. $\mathbb{Z}_2 \cong W_{n,d}$. Thus $W_{n,d}$ is perfectoid

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